

Lec 16: TCP congestion control

16/10/23

RECAP- * TCP 3-way handshake ... discuss FSM
* TCP retransmission scenarios.

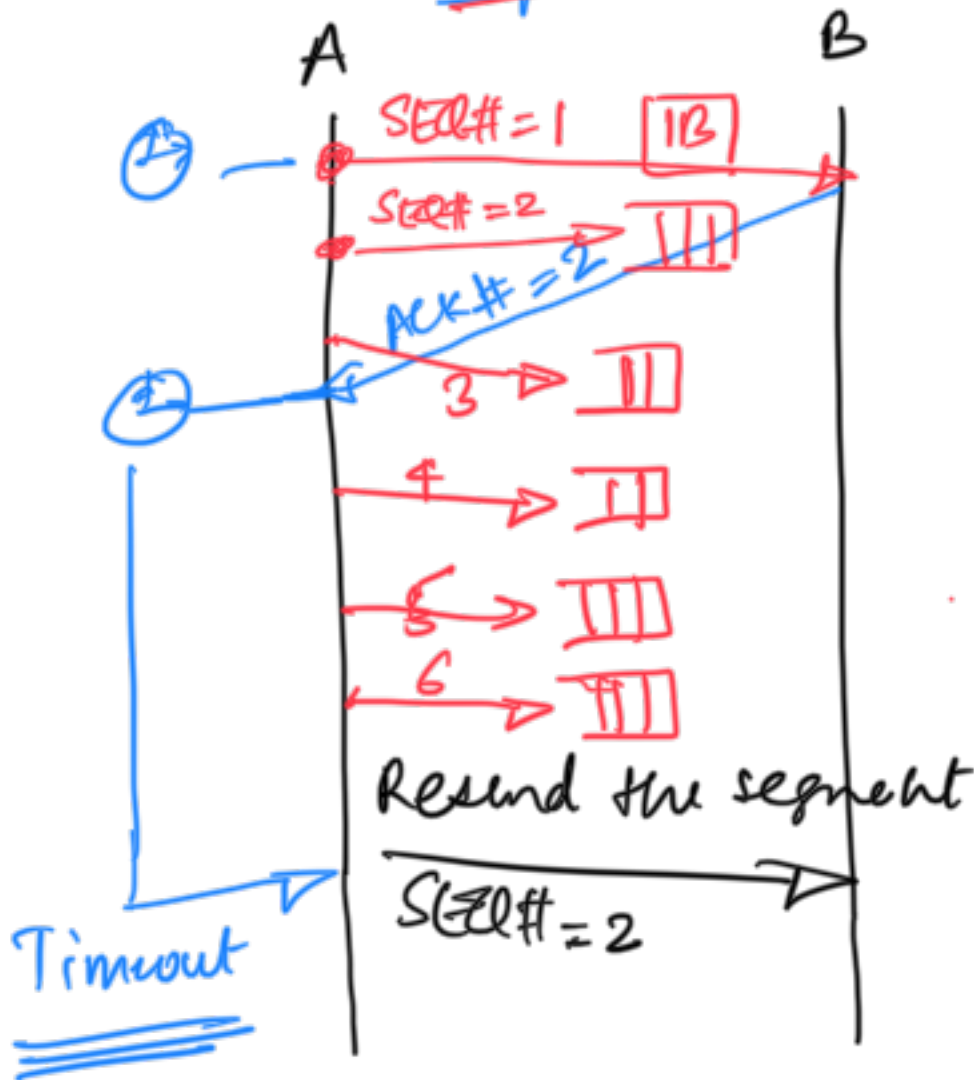


Refer slides

TCP congestion control

Q. How does TCP distinguish between CONGESTION & PACKET LOSS

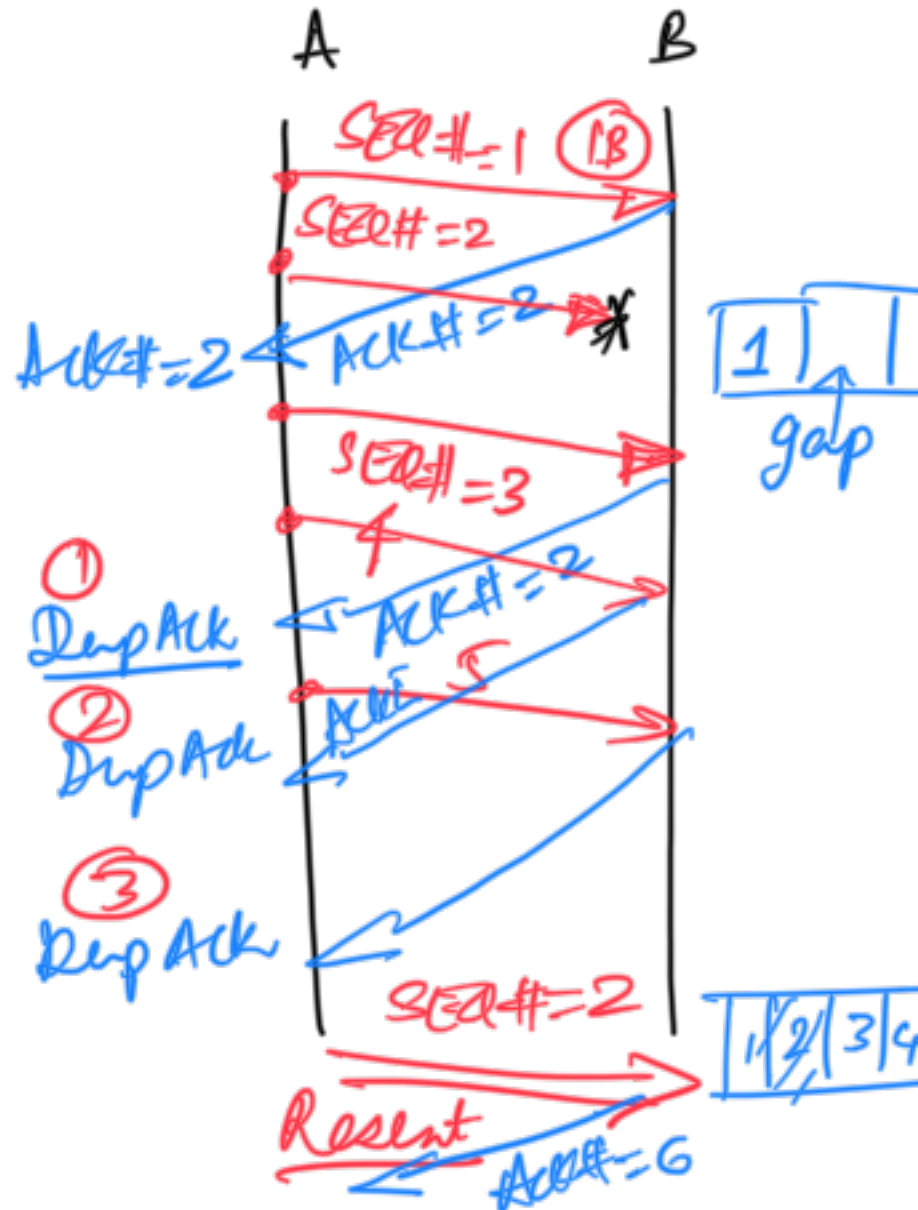
Congestion



+ Congestion
Congestion indicator
↳ Timeout

Solution: sender should slow down
* Slow start.

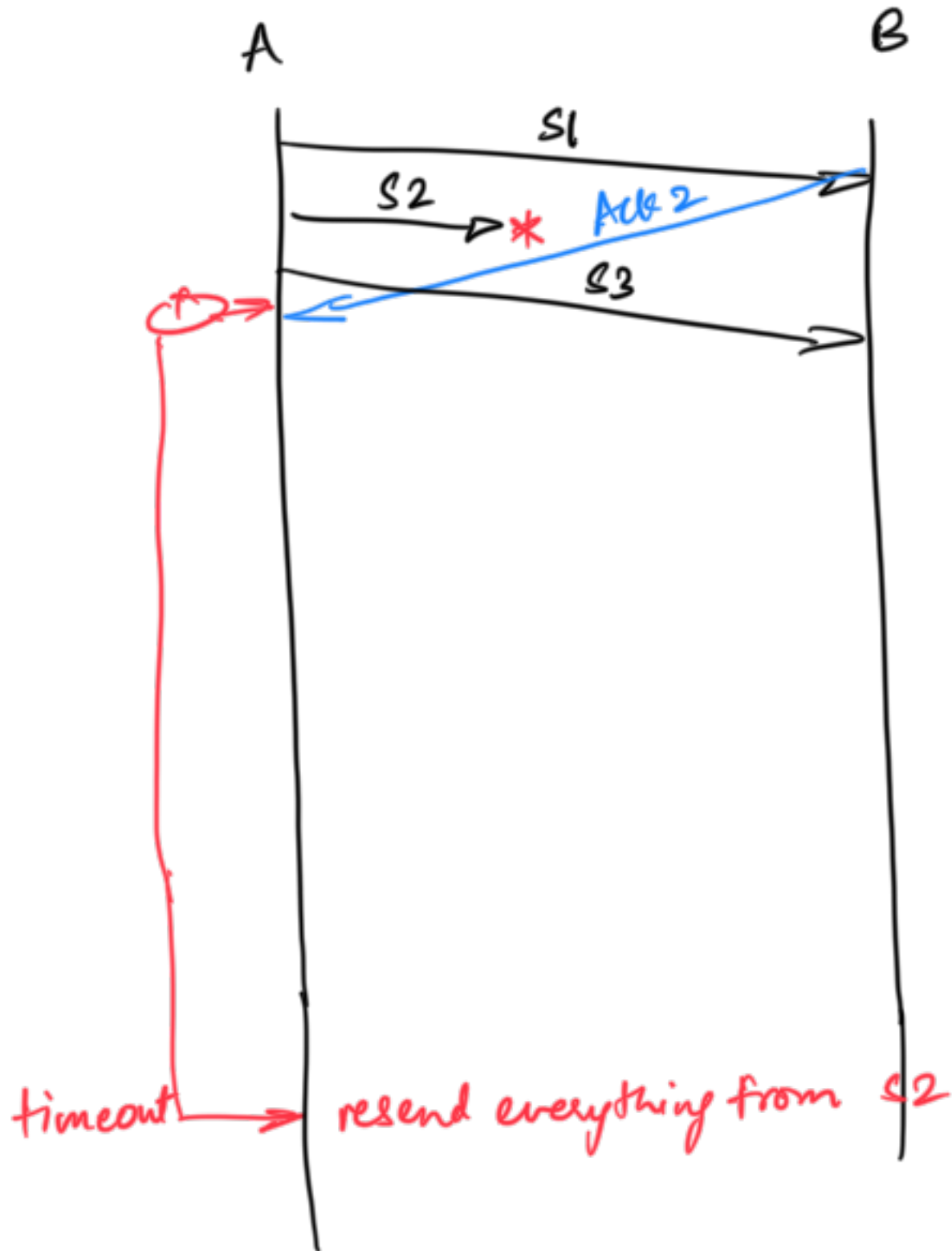
Packet loss



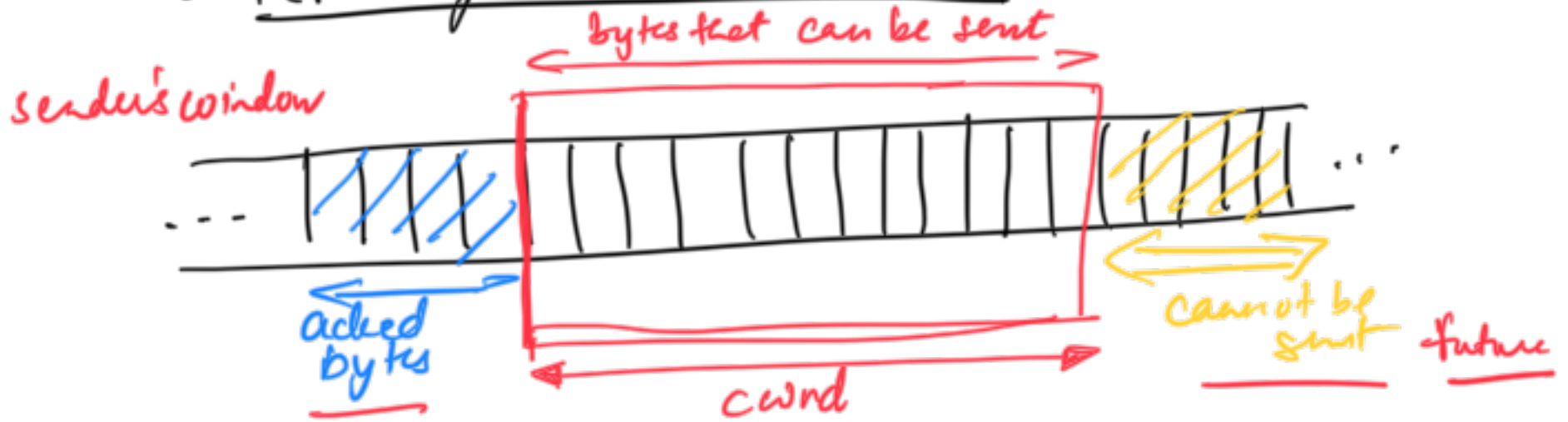
packet loss indicator
↳ 3 dup Ack

Solution: sender should quickly resend
* Fast retransmit/
just recovery

* TCP fast retransmit $\Rightarrow$ refer slides



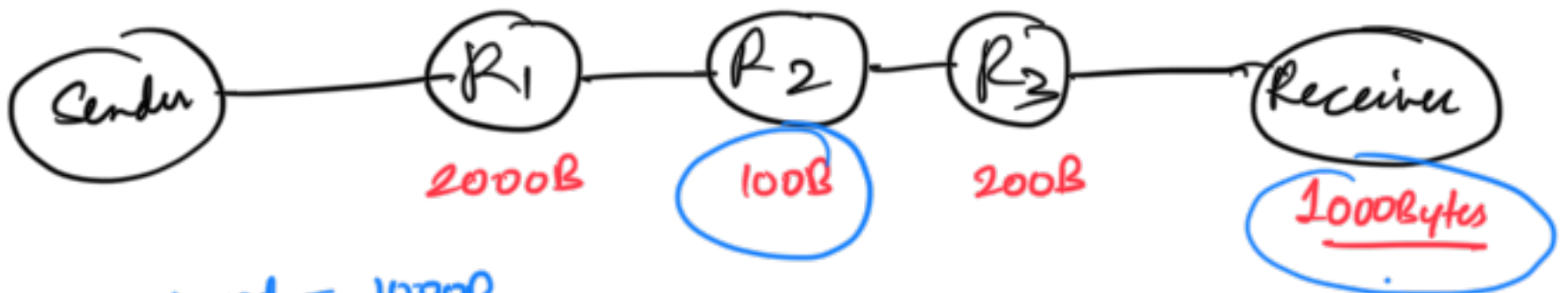
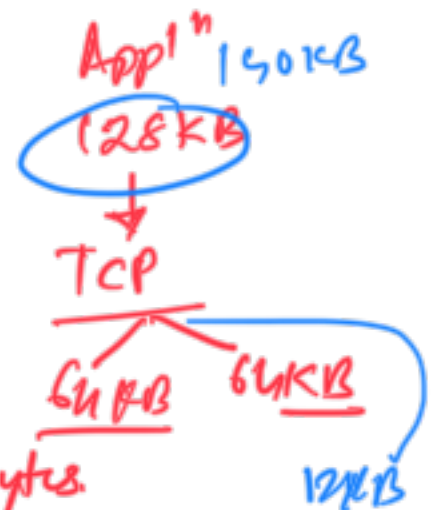
- * TCP congestion control
 - ↳ Causes/costs of congestion... refer slides
- TCP congestion control basics



- * Congestion window, cwnd in bytes
 - ↳ should refer to the bottleneck bandwidth

- * Window size = $\min(rwnd, cwnd)$
 - (sender should consider)

- * Negotiated option for a flow
 - ↳ Maximum segment size (MSS)
 - Default size = 64KB = 65536 Bytes.



$rwnd = 1000B$

Congestion control algo $\Rightarrow$ to estimate bottleneck BW
100B

* TCP congestion control : Slow start

TCP sender realizes congestion $\Rightarrow$ Timeout

$$W = 1 \text{ MSS} \quad \underline{\underline{64 \text{ KB}}}$$

If Ack; Double the window size

$$W = 2 \text{ MSS}$$

$$W = 4 \text{ MSS}$$

$$W = 8 \text{ MSS}$$

$$W = 16 \text{ MSS}$$

Timeout $\Rightarrow$ reduce the W to 1

$$W = 1 \text{ MSS}$$

